

ANALYSIS IN PROGRESS (status: 23 September 2026). Upstream processing is running one group at a time. Sections marked [PLANNED] describe the intended analysis and will be updated with final parameters, versions and results. THIS FILE SHOULD NOT BE USED FOR PUBLICATION AS IS.

Project

RNA-seq of mouse samples from an aging and intervention study (data: Roni Reggev). Four groups, three biological replicates each (12 samples): 2-month-old mice, 9-month-old control mice, 9-month-old mice treated with curcumin, and 9-month-old mice treated with plumbagin. Libraries were sequenced paired-end, 2×161 bp. Library preparation: SMARTer Stranded Total RNA-Seq Kit v3 – Pico Input Mammalian (Takara), i.e. ribosomal-RNA-depleted total RNA with unique molecular identifiers (UMIs) on Read 2 (the same kit as the lab's previous OB/OB project; to be confirmed with the sequencing core).

Processing overview

All samples are processed with the community-standard nf-core/rnaseq pipeline (release 3.26.0, run with Nextflow and Apptainer containers), with identical settings for every group. To limit disk use, the four groups are processed one after another; after each group, its final results are copied to D:\RONI REGGEV\RNA Seq Files\Final Results\<group> and verified before temporary files are removed. Processing groups separately does not introduce a batch effect, because software, reference and parameters are the same for all samples. Differential expression is then run once, on all 12 samples together.

Trimming and filtering of raw reads

Raw reads (fastq files) were inspected for quality issues with FastQC. The 8-nt UMI at the start of Read 2 was moved into the read name with UMI-tools extract, and the following 6 nt (3-nt UMI linker + 3 nt from the Pico v3 SMART UMI adapter) were removed, as recommended by the kit manufacturer. Reads were then trimmed with Trim Galore (a wrapper around cutadapt) using default settings: bases with quality below 20 were trimmed from the 3' end, Illumina adapter sequences were removed, and read pairs in which either read became shorter than 20 nt were discarded. Samples keeping fewer than 10,000 reads after trimming would be excluded from downstream steps.

Alignment and counting

The processed reads were aligned to the Mouse genome with STAR aligner. The reference genome version was GRCm39 (primary assembly), with annotations from Ensembl release 116. The STAR index was built with a splice-junction overhang of 160 (read length – 1). PCR duplicates were removed with UMI-tools dedup on the genome alignments: read pairs with the same mapping position and the same UMI are counted once, so remaining counts reflect original RNA molecules rather than PCR copies.

Gene-level counts ("Track A") were produced with featureCounts (Subread package) on the deduplicated alignments, at the Ensembl gene_id level, with strand information set to 'reverse'. Read pairs are counted as one fragment; only pairs with both reads aligned are counted, chimeric pairs are excluded, and reads mapping to more than one location or overlapping more than one gene are not counted. Nuclear and mitochondrial genes are counted together. Because multi-mapping reads are discarded, genes with close paralogs or pseudogenes may be under-counted; this should be kept in mind when interpreting such genes.

In addition, transcript abundances were estimated with Salmon (pseudo-alignment mode) as an independent, methodologically different cross-check. The pipeline's default gene-count route (Salmon on STAR transcriptome alignments) is replaced by the three quantification tracks described below.

Quantification strategy: Tracks A, B and C

Why three tracks. Two separate issues shape how reads are turned into numbers in this project. The first is technical. This kit captures total RNA after ribosomal-RNA depletion, and depletion is never complete, so a very large number of reads land on the small (16.3 kb) mitochondrial genome. In the lab's previous project with the same kit, mitochondrial read density was 1,300–2,000 times the genome average. UMI deduplication slows down much faster than linearly when that many reads pile up at the same positions. As a result, the pipeline's standard gene-count route (UMI deduplication on transcript alignments, then Salmon) could not finish, even when given 48 hours per sample. Deduplication on genome alignments is not affected, because it only processes each read's primary alignment. The second issue is biological. Measuring gene expression and measuring transposable elements (repeats such as L1) are different questions, and they need opposite handling of reads that match several places in the genome. Each track therefore answers one question in the way that suits it.

Track A: primary gene counts (used for differential expression; runs for every group now). What: featureCounts gene counts from the UMI-deduplicated genome alignments, as described above. Why: it uses the route that is not affected by the mitochondrial pile-up, it produces nuclear and mitochondrial gene counts together in one table, and it is fast and standard. Limitation: reads that match more than one location are discarded, not shared out. Genes with close paralogs, gene families or pseudogenes can therefore be under-counted, and this caveat is stated wherever Track A results are shown.

Track B: Salmon cross-check, nuclear genes only (optional; not yet scheduled). What: the 37 mitochondrial transcripts are removed from the STAR transcriptome alignments; UMI deduplication and Salmon quantification (alignment mode) are then run with the same arguments the pipeline would use. Why: removing the mitochondrial pile-up lets deduplication finish. Salmon then shares reads that match several genes or isoforms between them statistically instead of discarding them, which covers exactly the genes where Track A is weakest. Use: a cross-check for genes of interest that belong to gene families, not a replacement for Track A. Track B has no mitochondrial counts by design; those always come from Track A. Its run time is long and hard to predict. The STAR transcriptome alignments that Track B needs are kept for every sample (star_salmon/transcriptome_bam), so Track B can be run later without realigning.

Track C: transposable elements / L1 (planned; after Track A is complete for all groups). What: a separate STAR realignment that allows reads to match many locations (up to about 100), followed by TETranscripts (repeat-family level) and TELocal (individual-locus level), using a RepeatMasker repeat annotation for GRCm39, with its own normalization. Why: this asks a different biological question, namely whether aging loosens heterochromatin and reactivates young, potentially active L1 subfamilies (L1Md_A, L1Md_T, L1Md_Gf), and whether curcumin or plumbagin reverses this. It complements the lab's γ H2AX, H3K9me2, H3K27me3 and L1 ORF imaging. Copies of these repeats are nearly identical, so their reads match many places; the standard alignment used for Tracks A and B discards exactly those reads, which is why a separate realignment is needed rather than a setting change. The total-RNA kit helps here, because many repeat transcripts are not polyadenylated and would be missed by poly-A selection. Because low-input libraries can produce artifacts that look like repeat signal, 3' bias, internal priming and PCR duplication will be checked before any L1 result is reported. Track C starts again from the raw fastq files kept on D:, so it does not depend on any temporary files.

Quality control

Quality control is reported per sample and summarised in one MultiQC report per group. It includes: read quality (FastQC), alignment statistics (STAR, SAMtools), UMI deduplication rates (UMI-tools),

strandedness, read distribution and gene-body coverage for 3'/5' bias (RSeQC), RNA-seq QC (Qualimap), library complexity (Preseq), duplication versus expression (dupRadar), and ribosomal/mitochondrial content via biotype counts. No reads are removed in silico for rRNA; depletion efficiency is monitored in the QC instead.

Differential expression [PLANNED]

Normalization and differential expression analysis will be performed with the DESeq2 package on the Track A counts of all 12 samples together. The model is a single factor with four levels (design ~ group, plus a batch term if sample-collection or library batches are identified), with the 9-month control group (O9C) as the reference level. This is not a full 2×2 design: there are no treated 2-month animals, so an age $\times$ treatment interaction cannot be estimated, and each question is tested as a specific contrast (see Comparisons). Genes with at least 10 counts in at least 3 samples will be kept. Variance-stabilized (VST) counts will be used for quality control plots such as sample-distance heatmaps and principal component analysis. Criteria for excluding outlier samples will be fixed before looking at differential expression results. Log2 fold changes will be shrunk with the ash method, and genes with an adjusted p-value (Benjamini–Hochberg) below 0.05 will be called significant; any fold-change cut-off used for plots will be reported separately. With $n = 3$ per group, statistical power is limited, and this will be stated wherever results are shown.

Rescue analysis: genes changed with age (O9C vs Y2) will be classified for each treatment as rescued (significant change in the opposite direction), partially rescued (opposite direction, not significant), unchanged, or exacerbated (same direction as aging). Pathway analysis will use gene set enrichment (fgsea / clusterProfiler) on GO Biological Process, KEGG/Reactome and MSigDB Hallmark gene sets, with focused sets for senescence/SASP, inflammation, oxidative stress/Nrf2, mitochondrial/OXPHOS and heterochromatin/epigenetic regulators.

Abbreviations:

Y2: Young mice, 2 months old, untreated. Samples Y2_1–3 = raw files Roni_2MR1–3 (S40–S42).

O9C: Aged mice, 9 months old, control. Samples O9C_1–3 = raw files Roni_9MR1–3 (S43–S45).

O9Cur: Aged mice, 9 months old, treated with curcumin. Samples O9Cur_1–3 = raw files Roni_CURCAR1–3 (S46–S48).

O9Pul: Aged mice, 9 months old, treated with plumbagin. Samples O9Pul_1–3 = raw files Roni_PLUMR1–3 (S49–S51).

UMI: unique molecular identifier; **Track A / B / C:** primary gene counts / Salmon cross-check / transposable elements (see Quantification strategy); **padj:** Benjamini–Hochberg adjusted p-value;

LFC: log2 fold change.

Comparisons:

C1_Aging = O9C vs. Y2 (what changes with age)

C2_Curcumin = O9Cur vs. O9C (effect of curcumin in aged mice)

C3_Plumbagin = O9Pul vs. O9C (effect of plumbagin in aged mice)

C4_CurVsPlum = O9Cur vs. O9Pul (difference between the two treatments)

C5_Cur_Young / C5_Plum_Young = O9Cur vs. Y2 and O9Pul vs. Y2 (remaining distance from young after treatment)

Rescue = derived from C1 together with C2 / C3 (does treatment move age-changed genes back toward young levels)

Result files and naming conventions:

Final results for each group are stored in D:\RONI REGGEV\RNA Seq Files\Final Results\<group> (Y2, O9C, O9Cur, O9Pul):

- track_a/<sample>.gene_counts.tsv = Track A gene counts (the input for differential expression); .summary = how many read pairs were assigned or why they were not.
- star_salmon/<sample>.umi_dedup.sorted.bam (+ .bai) = deduplicated genome alignments, viewable in IGV.
- star_salmon/bigwig = strand-specific coverage tracks for genome browsers.
- star_salmon/transcriptome_bam/<sample>.Aligned.toTranscriptome.out.bam = STAR alignments to transcripts (input for Track B; not deduplicated).
- star_salmon/rseqc, qualimap, preseq, dupradar, samtools_stats = per-sample QC outputs.
- salmon = Salmon pseudo-alignment quantification (cross-check).
- multiqc = one combined, interactive QC report per group (multiqc_report.html); the easiest place to start.
- fastqc, trimgalore, umitools = read-level QC, trimming and UMI logs.
- pipeline_info = exact software versions and run parameters.
- samplesheet_<group>.csv = mapping of sample names to raw fastq files.

[PLANNED] Differential expression outputs will be provided per comparison: a full results table (Ensembl ID, gene symbol, biotype, baseMean, LFC, shrunk LFC, p-value, padj, raw and normalized counts), MA and volcano plots (grey = non-significant, red = significant genes), and heatmaps of significant genes. File-naming conventions for plots will be added here when those files are produced.

Missing values:

In DESeq2 results, a gene can get 'NA' values: if both pvalue and padj are NA, the gene has a sample with an extreme count compared with the other samples (a Cook's distance outlier) and was not tested; if only padj is NA, the gene was removed by DESeq2's independent filtering because its expression was too low to reach significance; genes with zero counts in all samples have NA in all statistics columns. Genes removed by the pre-filter (see above) have raw counts but no normalized counts or statistics.

Citations

nf-core

Ewels PA, Peltzer A, Fillinger S, Patel H, Alneberg J, Wilm A, Garcia MU, Di Tommaso P, Nahnsen S. The nf-core framework for community-curated bioinformatics pipelines. Nat Biotechnol. 2020;38(3):276-278. doi:10.1038/s41587-020-0439-x

Nextflow

Di Tommaso P, Chatzou M, Floden EW, Barja PP, Palumbo E, Notredame C. Nextflow enables reproducible computational workflows. Nat Biotechnol. 2017;35(4):316-319. doi:10.1038/nbt.3820

UMI-tools

Smith T, Heger A, Sudbery I. UMI-tools: modeling sequencing errors in Unique Molecular Identifiers to improve quantification accuracy. Genome Res. 2017;27(3):491-499. doi:10.1101/gr.209601.116

Cutadapt

Martin M. Cutadapt removes adapter sequences from high-throughput sequencing reads. EMBnet.journal 2011, 17.1:10-12. doi:10.14806/ej.17.1.200

STAR

Dobin A, Davis CA, Schlesinger F, Drenkow J, Zaleski C, Jha S, Batut P, Chaisson M, Gingeras TR. STAR: ultrafast universal RNA-seq aligner. Bioinformatics. 2013;29(1):15-21. doi:10.1093/bioinformatics/bts635

SAMtools

Danecek P, Bonfield JK, Liddle J, Marshall J, Ohan V, Pollard MO, Whitwham A, Keane T, McCarthy SA, Davies RM, Li H. Twelve years of SAMtools and BCFtools. GigaScience. 2021;10(2):giab008. doi:10.1093/gigascience/giab008

featureCounts

Liao Y, Smyth GK, Shi W. featureCounts: an efficient general purpose program for assigning sequence reads to genomic features. *Bioinformatics*. 2014;30(7):923-930. doi:10.1093/bioinformatics/btt656

Salmon

Patro R, Duggal G, Love MI, Irizarry RA, Kingsford C. Salmon provides fast and bias-aware quantification of transcript expression. *Nat Methods*. 2017;14(4):417-419. doi:10.1038/nmeth.4197

RSeQC

Wang L, Wang S, Li W. RSeQC: quality control of RNA-seq experiments. *Bioinformatics*. 2012;28(16):2184-2185. doi:10.1093/bioinformatics/bts356

MultiQC

Ewels P, Magnusson M, Lundin S, Källér M. MultiQC: summarize analysis results for multiple tools and samples in a single report. *Bioinformatics*. 2016;32(19):3047-3048. doi:10.1093/bioinformatics/btw354

DESeq2

Love MI, Huber W, Anders S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology* 2014, 15:550. doi:10.1186/s13059-014-0550-8

ashr

Stephens M. False discovery rates: a new deal. *Biostatistics*. 2017;18(2):275-294. doi:10.1093/biostatistics/kxw041

Program versions

Versions as pinned by nf-core/rnaseq 3.26.0; the exact versions used are recorded in pipeline_info for each group.

nf-core/rnaseq:	3.26.0, https://nf-co.re/rnaseq
Nextflow:	25.04.7
FastQC:	v0.12.1
UMI-tools:	v1.1.6 (extract, dedup)
Trim Galore:	v2.1.0 (cutadapt wrapper)
STAR:	v2.7.11b
SAMtools:	v1.21 / v1.23.1
featureCounts	v2.0.6
(Subread):	
Salmon:	v1.10.3
RSeQC:	v5.0.4
Qualimap:	v2.3
Preseq:	v3.2.0
dupRadar:	v1.38.0
MultiQC:	v1.33

Reference: Mus musculus GRCm39 primary assembly, Ensembl release 116

R / DESeq2: [PLANNED] R 4.4 with DESeq2, ashR, pheatmap, ggplot2, ggrepel;
versions to be recorded at analysis